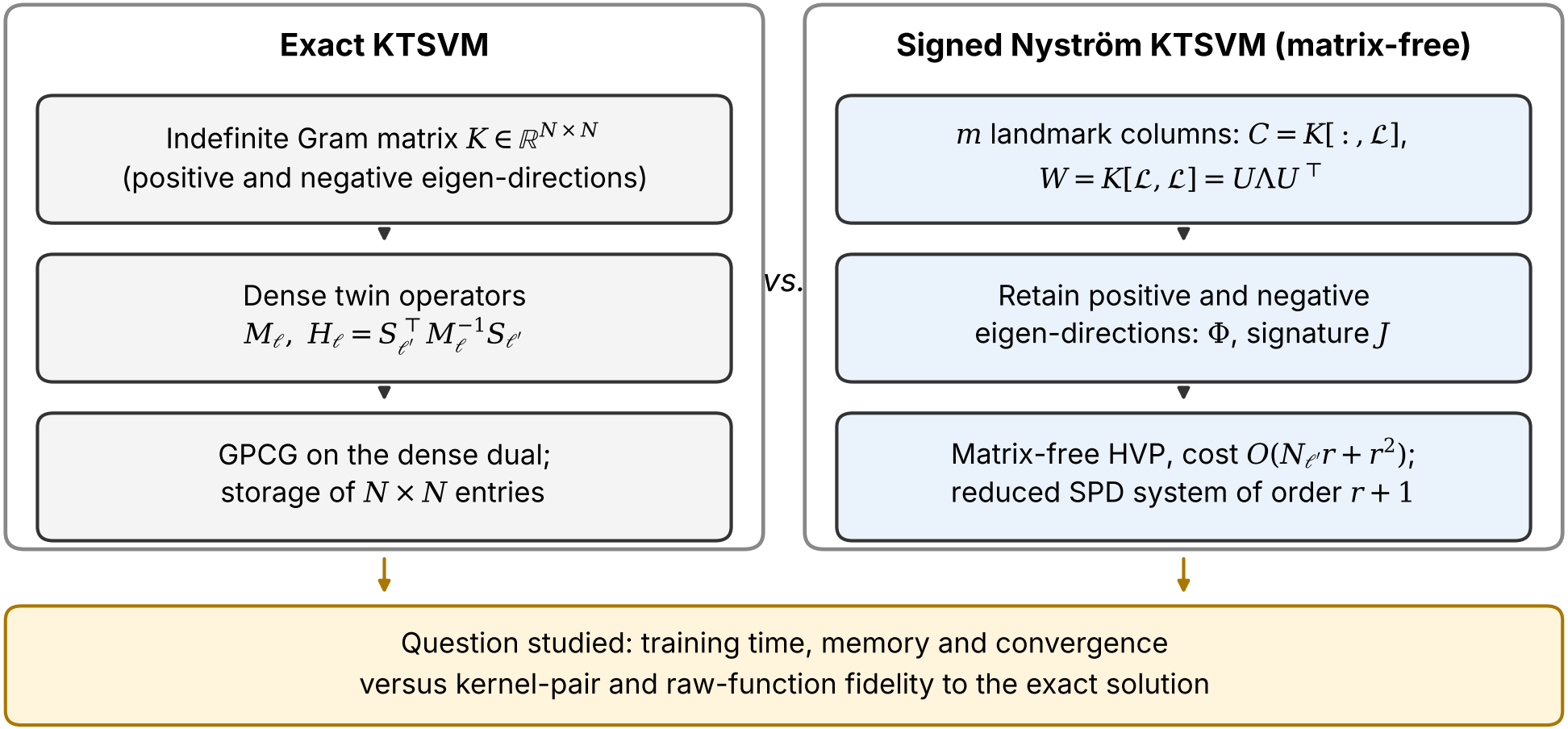
Exact KTSVM
Indefinite Gram matrix $K \in \mathbb{R}^{N \times N}$
(positive and negative eigen-directions)
Dense twin operators
$M_\ell,\ H_\ell = S_{\ell'}^\top M_\ell^{-1} S_{\ell'}$
GPCG on the dense dual;
storage of $N \times N$ entries
vs.
Signed Nyström KTSVM (matrix-free)
m landmark columns: $C = K[:, \mathcal{L}]$,
$W = K[\mathcal{L}, \mathcal{L}] = U \Lambda U^\top$
Retain positive and negative
eigen-directions: Φ, signature J
Matrix-free HVP, cost $O(N_{\ell'} r + r^2)$;
reduced SPD system of order $r + 1$
Question studied: training time, memory and convergence
versus kernel-pair and raw-function fidelity to the exact solution